

Sujal Sirimilla

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Education

UNIVERSITY OF CALIFORNIA, SAN DIEGO

B.S in Data Science | Electrical Engineering Minor + Mathematics Minor

GPA: 3.7 (Provost Honors)

Class of 2028

(Junior/3rd Year standing)

Experience

Duarte Lab at UCSD

Jan 2026 – Present

Undergraduate Researcher

- Conduct research on interpretable machine learning for jet tagging using Generalized Additive Models (GAMs) and baseline ML methods.
- Implement data preprocessing, feature analysis, and model evaluation pipelines in Python.
- Analyze model interpretability and performance to understand physically meaningful patterns in particle jet data.

TRITON ROBOTICS

Jan 2026 – Present

Webmaster

- Selected for the Webmaster team of Triton Robotics, a competitive UCSD robotics organization.
- Contribute to the design, development, and maintenance of the club's official website, ensuring a reliable and engaging platform for members and external audiences. Redesigning a new website (React+Netlify)
- Building Interactive 3D robot simulator with real-time keyboard teleoperation using **Three.js** and WebGL.

UCSD MATHEMATICS

Sep 2025 – Present

Tutor at ASC@Math

- Provide one-on-one & small-group tutoring in calculus and precalculus, supporting >50 students weekly.
- Adapt explanations to diverse learning styles, help students clarify concepts and work through problems.
- Help students develop problem-solving strategies & study approaches. Give personalized exam feedback.
- Foster academic environment where students feel comfortable asking questions & engage with material.

Projects

WEBSITE FOR WAIT TIMES (github.com/ssirimilla/airport-pred-flask)

Dec 2025 – Jan 2026

- Trained & deployed an XGBoost regression model for airport wait-time prediction with cyclical time features. Compared multiple baselines (mean predictor, linear regression, random forest).
- Performed fairness evaluation across citizenship groups & documented persistent error disparities.
- Used git for version control and Packaged the model behind a Flask API and deployed to a cloud platform (Render) for scalable inference.

PREDICTING POWER OUTAGES (github.com/ssirimilla/power-analysis)

Sep 2025 – Dec 2025

- Created an end-to-end Machine Learning pipeline predicting Hurricane-related power outage severities across US. Used Scikit-Learn BinaryClassifier with RandomizedSearchCV to find salient hyperparameters.
- Performed data cleaning, EDA, hypothesis testing, and feature engineering using Pandas, Designed & deployed a static portfolio website with plotly, matplotlib, plotly graph objects & Power BI visualizations.
- Conducted fairness evaluations using permutation tests & bias analysis. Model achieves 98% accuracy.

Skills & Coursework

- **Programming & Scientific Computing:** Python, NumPy, SciPy, Pandas, MATLAB
- **MachineLearning & Stats:** Sklearn, XGBoost, FeatureEngineering, ModelEvaluation, Fair/Bias Analysis
- **Data Visualization & BI:** Matplotlib, Plotly, Folium, Power BI
- **Tools & Platforms:** Git/GitHub, JupyterLab, Google Colab, Render, Flask, Jekyll, GitHub Pages
- **Data Science (DSC10-100):** Principles of Data Science, Programming & DSA, Data Management (SQL)
- **Mathematics (20Series, Math 183) :** Calculus (full sequence), Linear Algebra, Probability + Statistics
- **Electrical & Computer Engineering (ECE35,45):** Introduction to Analog Design, Circuits and Systems